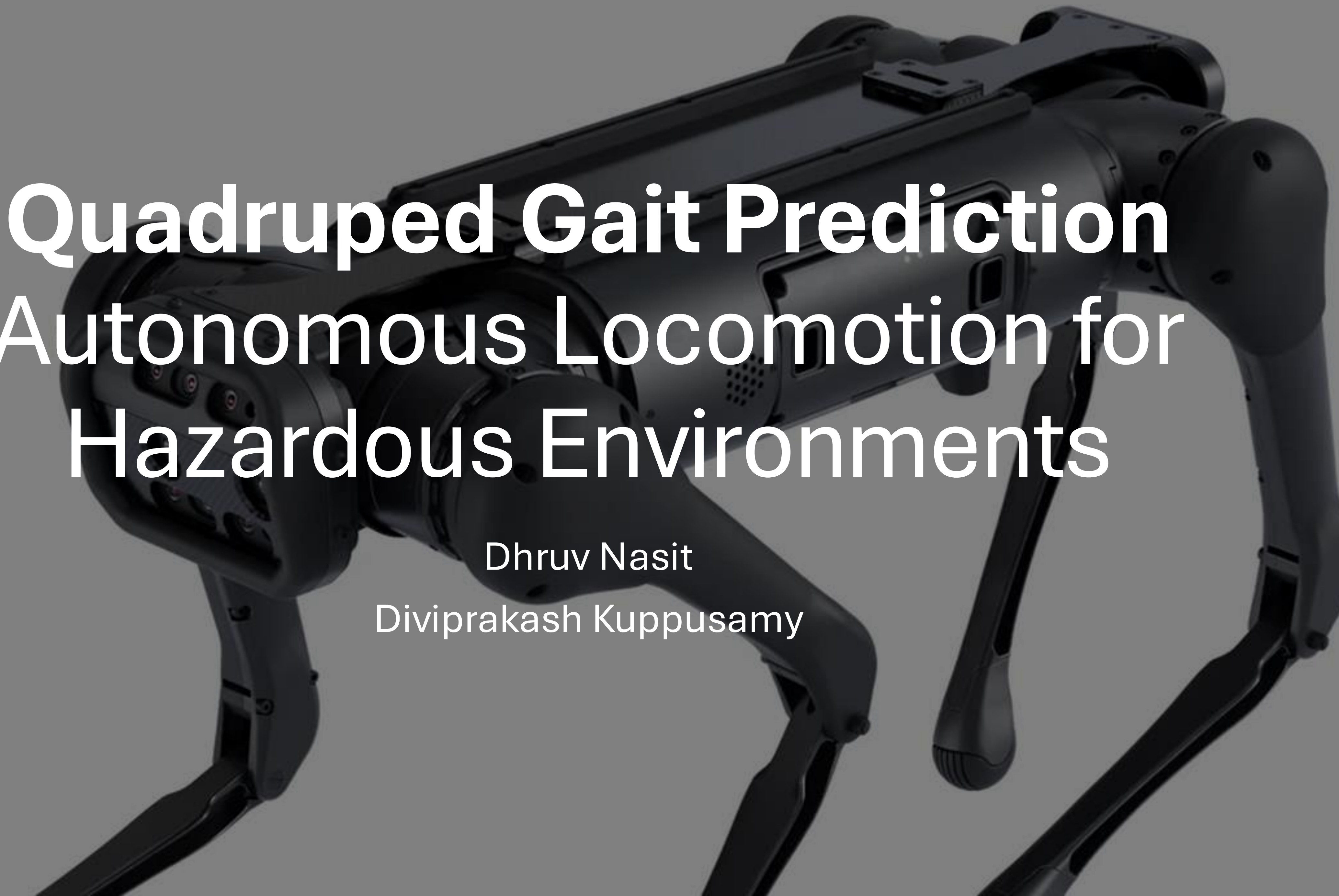

Quadruped Gait Prediction
Autonomous Locomotion for Hazardous Environments

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The Problem

Construction Work is High Risk

- One of the highest workplace injury and fatality rates
- Falls, unstable terrain, structural collapse
- Hazard inspections still rely on human exposure
- Small firms lack advanced safety automation

Key Question: How can we reduce human exposure in dangerous environments?

Why This Matters Now

Rapid growth in robotics & automation

Advances in: Physics-based simulation, Reinforcement learning, Autonomous control systems

Increased safety expectations across industries

Simulation tools now allow safe development before deployment.

Project Purpose

Design an intelligent autonomous system that:

- Navigates hazardous terrain
- Operates without human intervention
- Reduces safety risk
- Improves situational assessment

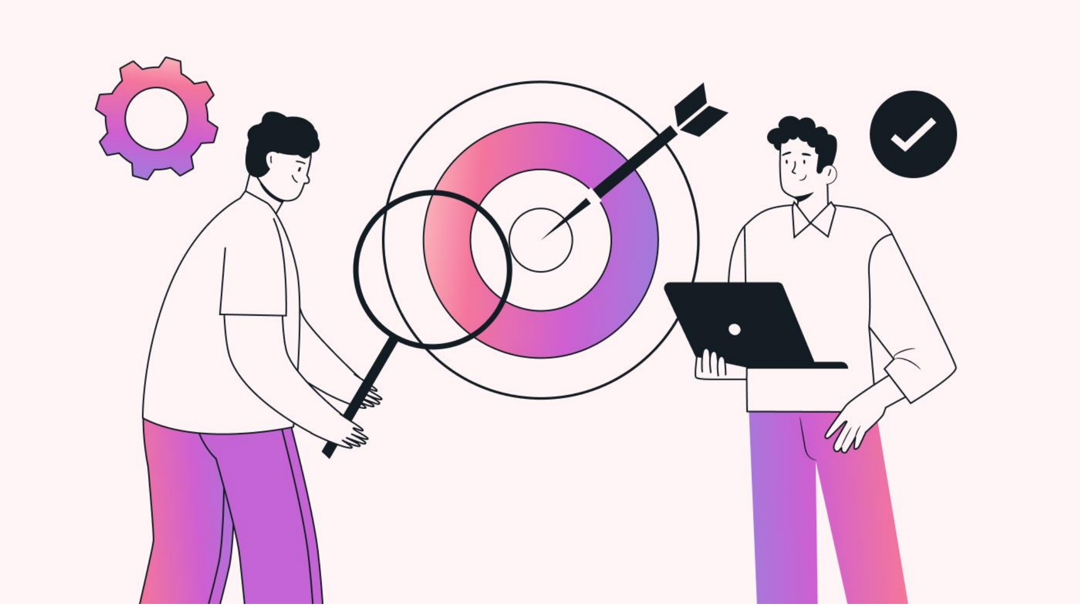
End Result:

- A validated autonomous locomotion system tested in simulation.

Objectives

1. Create a simulated hazardous terrain environment
2. Develop a machine learning-based locomotion model
3. Enable stable autonomous navigation
4. Evaluate system performance using measurable metrics

- **Success = Stable, consistent traversal without human control**



System Architecture

- Hazard Simulation
- State Inputs (terrain, stability, orientation)
- ML-Based Control Model
- Autonomous Gait Output
- Performance Evaluation

Methodology: Simulation

Simulation Environment:

- Physics-based hazardous terrain
- Unstable surfaces & obstacles
- Controlled test scenarios
- Repeatable experiments

Note: Training data generated entirely from simulations.

Methodology: Model

Locomotion Model

- Machine learning-based control system
- Trained using simulation data
- Autonomous movement without human input

Focus Areas:

- Stability
- Balance
- Adaptive gait control

Evaluation

We measure:

- Traversal Success Rate
- Stability Score
- Failure Frequency
- Consistency Across Runs

Goal: Reliable navigation across hazardous terrain.

Constraints & Risks

Constraints:

- 11-week academic timeline
- Simulation-only environment
- Limited computing resources

Risks:

- Model may not generalize to all terrains
- Simulation ≠ Real-world conditions
- Limited time for tuning

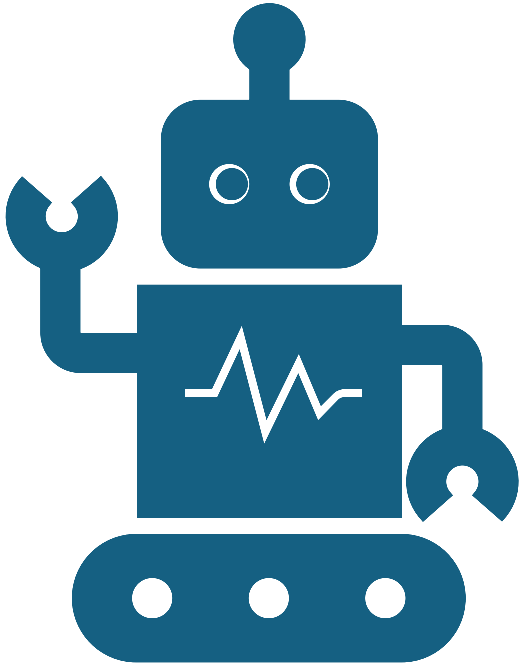
Future Work

Short-Term:

- Improve locomotion robustness
- Expand hazardous scenarios
- Optimize training

Long-Term Vision:

- Physical robot deployment
- Real-world validation
- Industry integration



Conclusion

- Reduces human exposure in hazardous environments
- Demonstrates practical ML application
- Provides scalable safety-focused robotics framework
- Safer environments through intelligent locomotion.